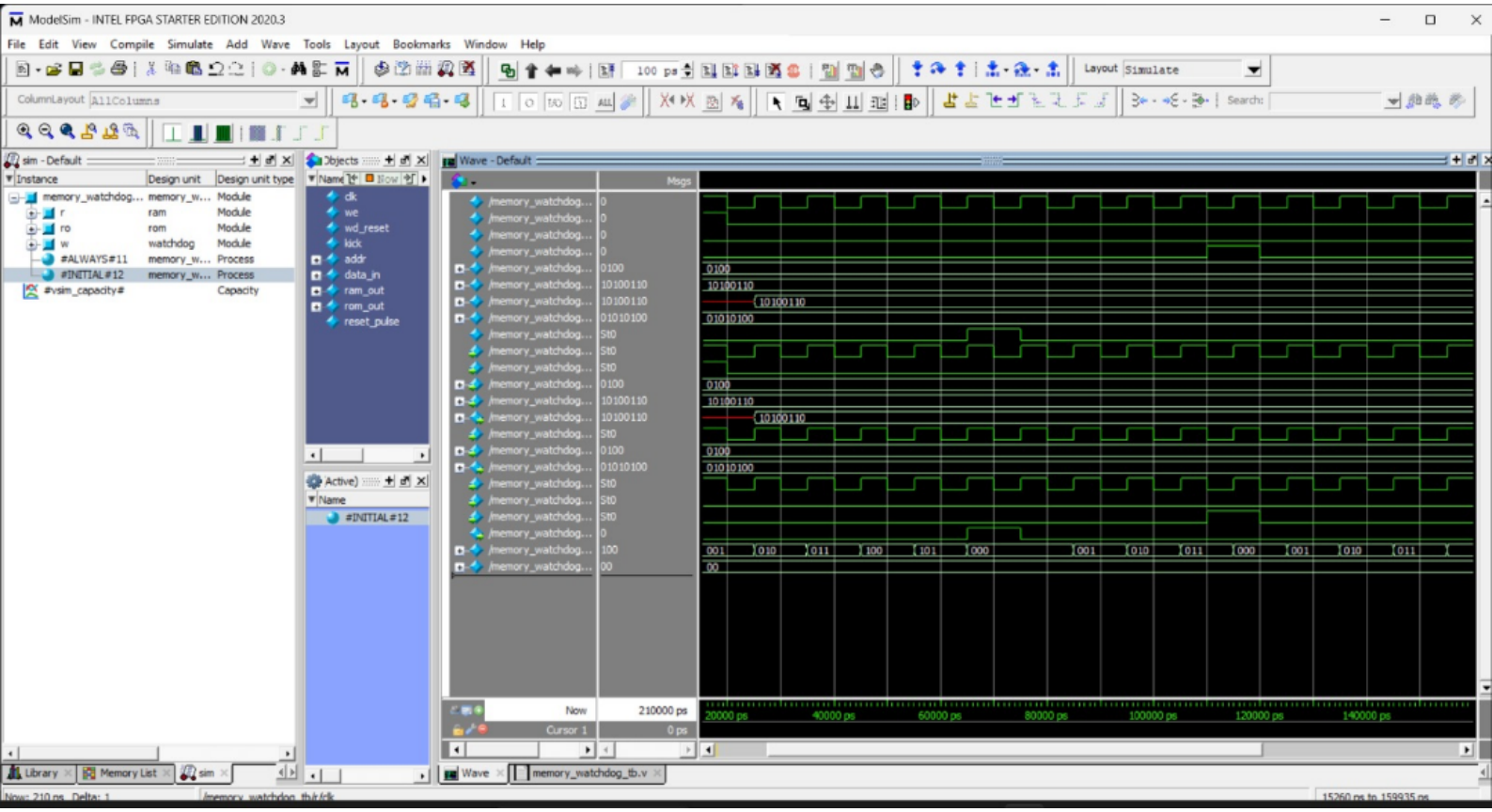
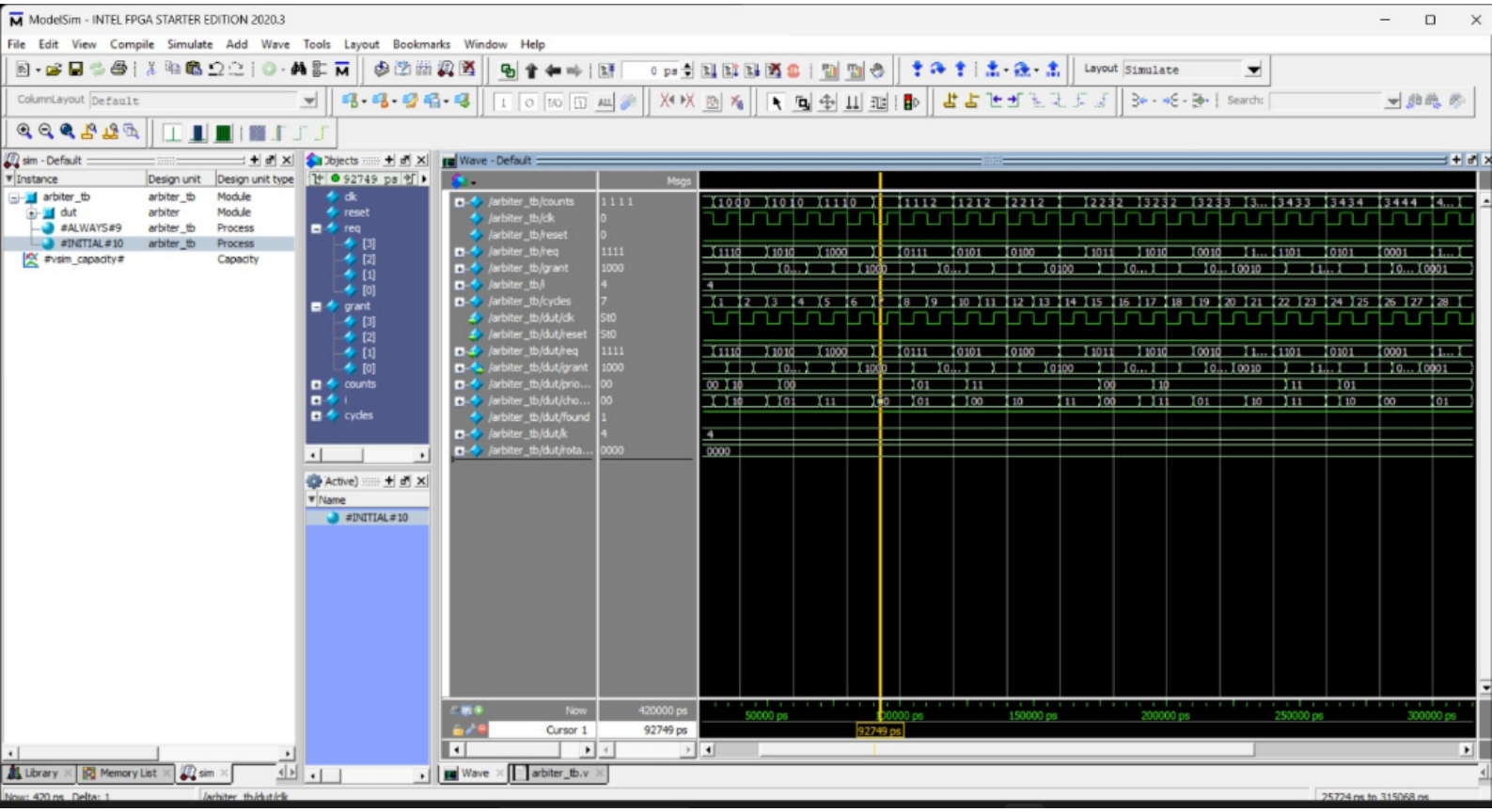
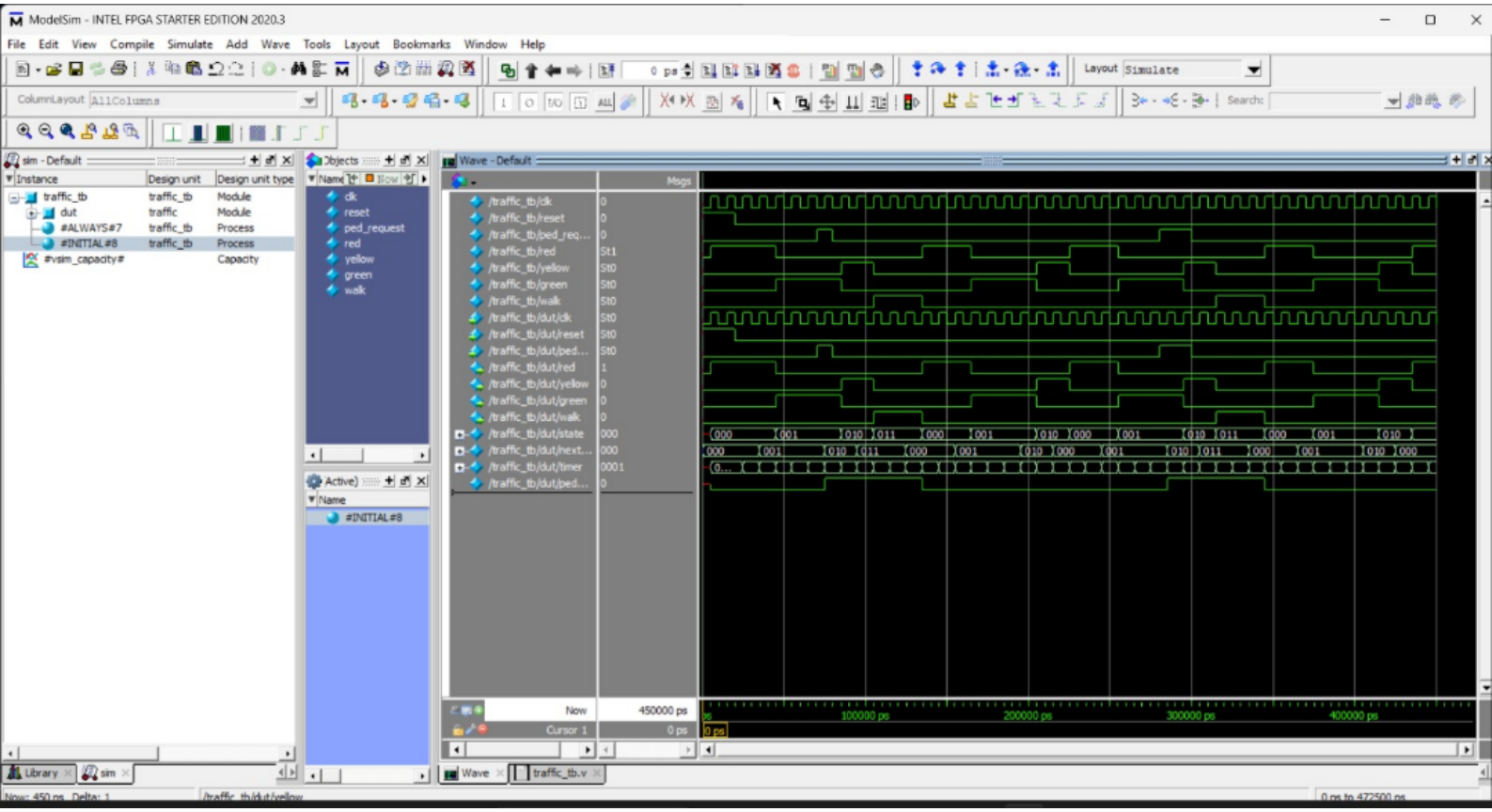
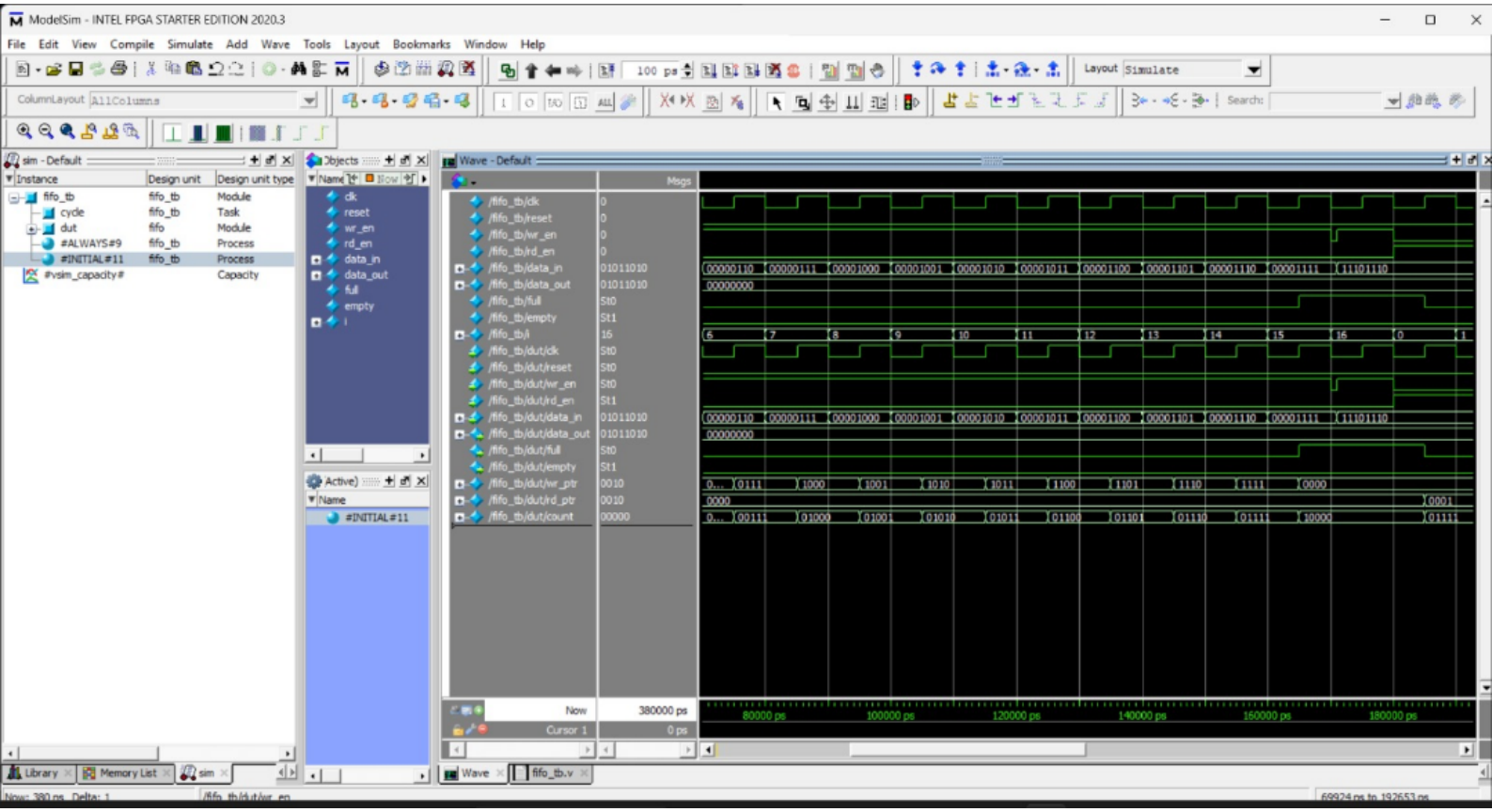
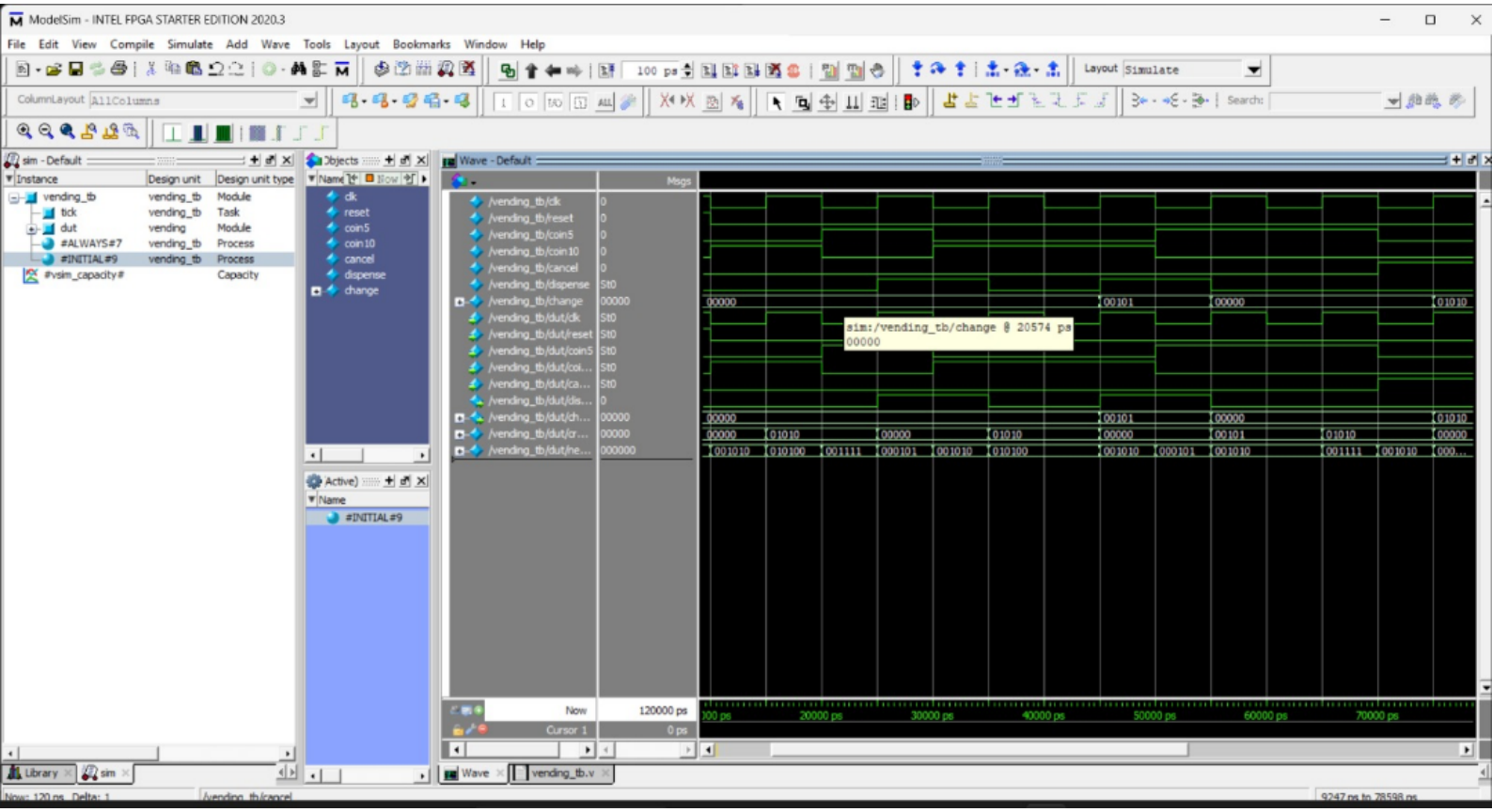
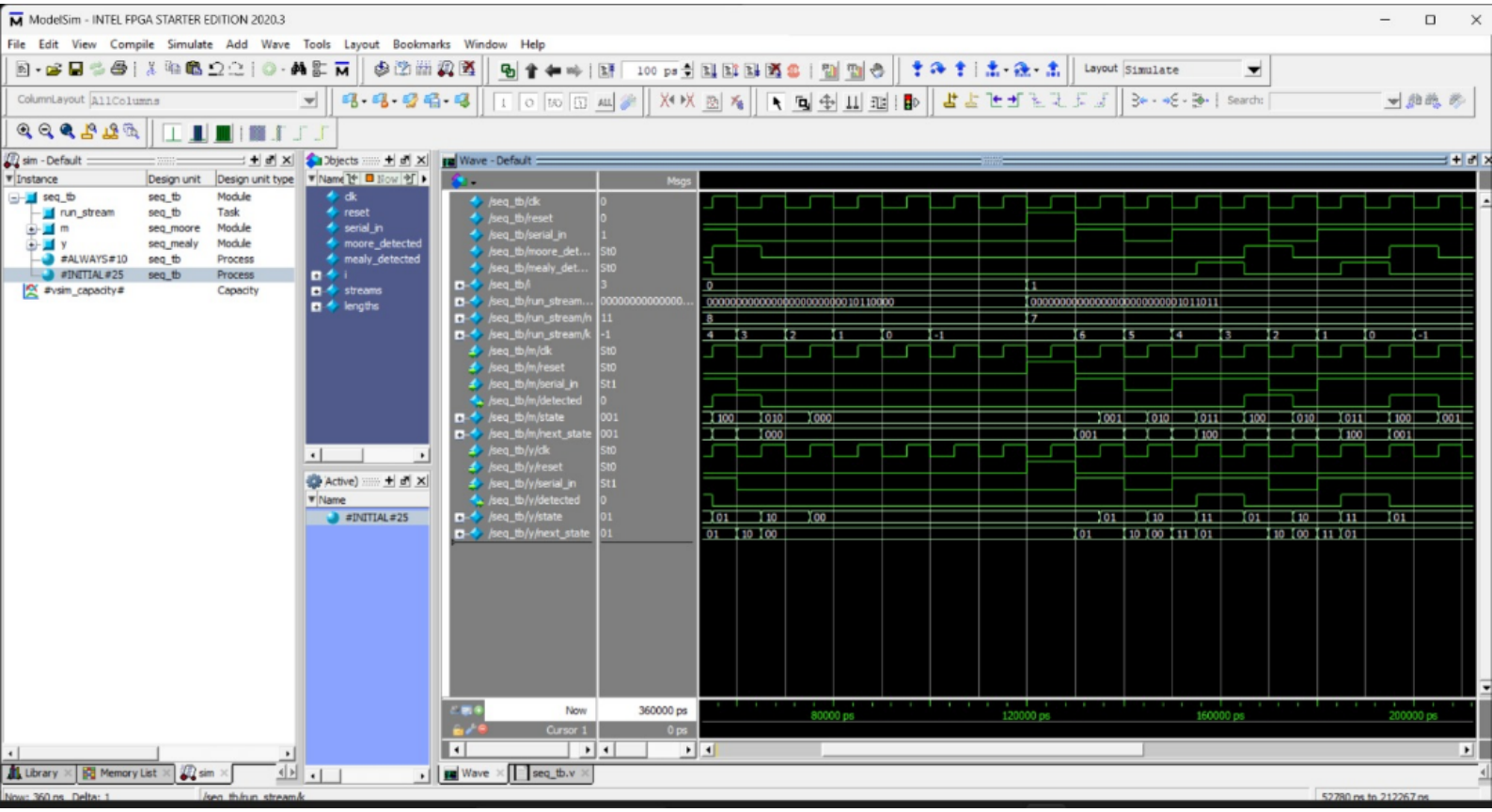

ModelSim - INTEL FPGA STARTER EDITION 2020.3
File Edit View Compile Simulate Add Wave Tools Layout Bookmarks Window Help
sim - Default
Instance
Design unit
Design unit type
memory_watchdog...
memory_w...
Module
r
ram
Module
ro
rom
Module
w
watchdog
Module
#ALWAYS#11
memory_w...
Process
#INITIAL#12
memory_w...
Process
#vsim_capacity#
Capacity
Objects
clk
we
wd_reset
kick
addr
data_in
ram_out
rom_out
reset_pulse
Active)
Name
#INITIAL#12
Wave - Default
Msgs
/memory_watchdog...
0
0100
10100110
01010100
St0
100
001
010
011
000
00
Now
210000 ps
Cursor 1
0 ps
20000 ps
40000 ps
60000 ps
80000 ps
100000 ps
120000 ps
140000 ps
Library
Memory List
sim
Wave
memory_watchdog_tb.v

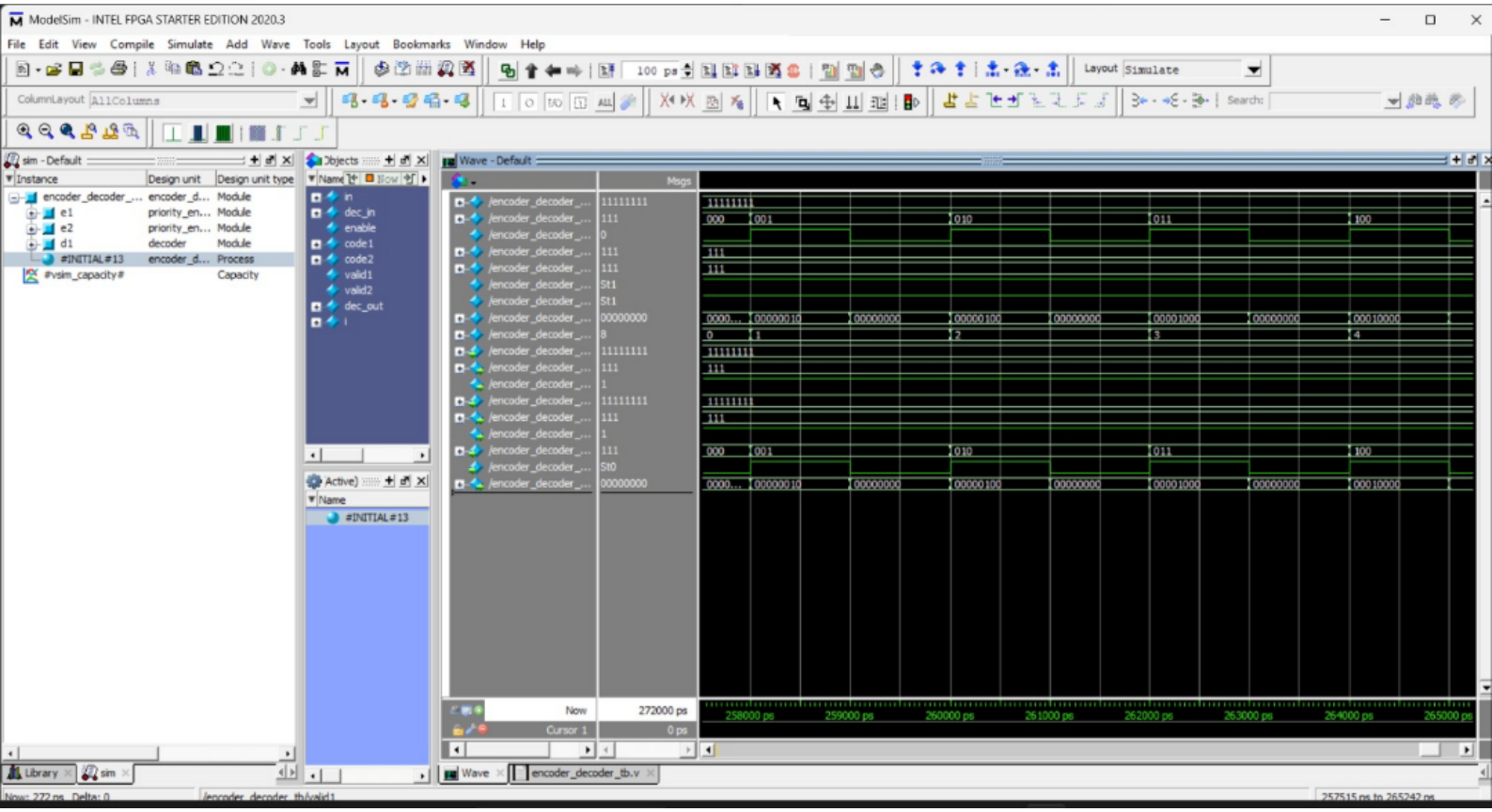

ModelSim - INTEL FPGA STARTER EDITION 2020.3
File Edit View Compile Simulate Add Wave Tools Layout Bookmarks Window Help
sim - Default
Instance
Design unit
Design unit type
arbiter_tb
dut
#ALWAYS#9
#INITIAL#10
#vsim_capacity#
Objects
clk
reset
req
grant
counts
i
cycles
Wave - Default
Msgs
/arbiter_tb/counts
/arbiter_tb/clk
/arbiter_tb/reset
/arbiter_tb/req
/arbiter_tb/grant
/arbiter_tb/i
/arbiter_tb/cycles
/arbiter_tb/dut/clk
/arbiter_tb/dut/reset
/arbiter_tb/dut/req
/arbiter_tb/dut/grant
/arbiter_tb/dut/found
/arbiter_tb/dut/k
Now
420000 ps
Cursor 1
92749 ps
50000 ps
100000 ps
150000 ps
200000 ps
250000 ps
300000 ps
Library
Memory List
sim
Wave
arbiter_tb.v

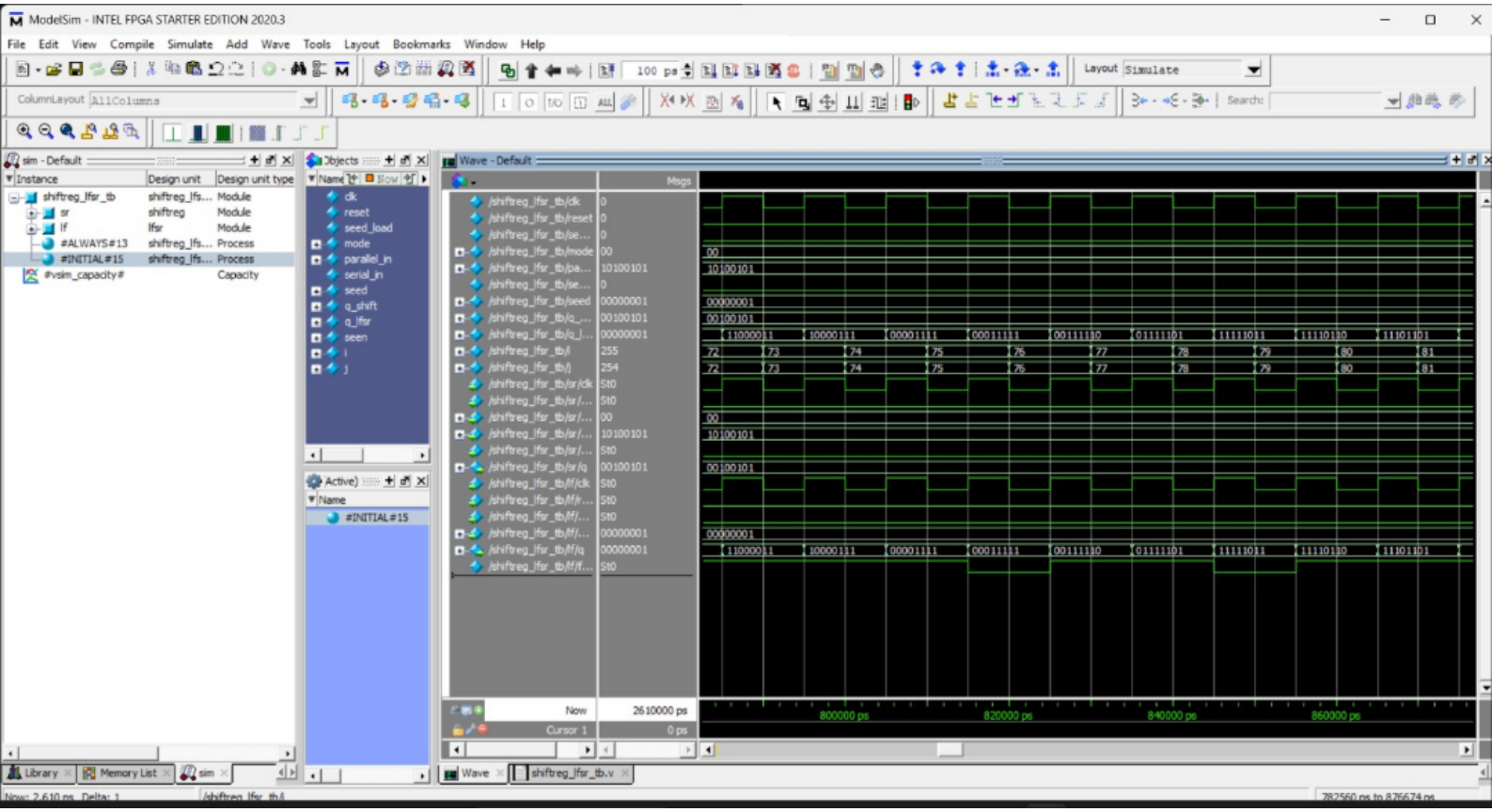

ModelSim - INTEL FPGA STARTER EDITION 2020.3
File Edit View Compile Simulate Add Wave Tools Layout Bookmarks Window Help
sim - Default
Instance
Design unit
Design unit type
traffic_tb
traffic_tb
Module
dut
traffic
Module
#ALWAYS#7
traffic_tb
Process
#INITIAL#8
traffic_tb
Process
#vsim_capacity#
Capacity
Objects
clk
reset
ped_request
red
yellow
green
walk
Active)
Name
#INITIAL#8
Wave - Default
Msgs
/traffic_tb/clk
/traffic_tb/reset
/traffic_tb/ped_req...
/traffic_tb/red
/traffic_tb/yellow
/traffic_tb/green
/traffic_tb/walk
/traffic_tb/dut/clk
/traffic_tb/dut/reset
/traffic_tb/dut/ped...
/traffic_tb/dut/red
/traffic_tb/dut/yellow
/traffic_tb/dut/green
/traffic_tb/dut/walk
/traffic_tb/dut/state
/traffic_tb/dut/next...
/traffic_tb/dut/timer
/traffic_tb/dut/ped...
Now
450000 ps
Cursor 1
0 ps
100000 ps
200000 ps
300000 ps
400000 ps
Library
sim
Wave
traffic_tb.v

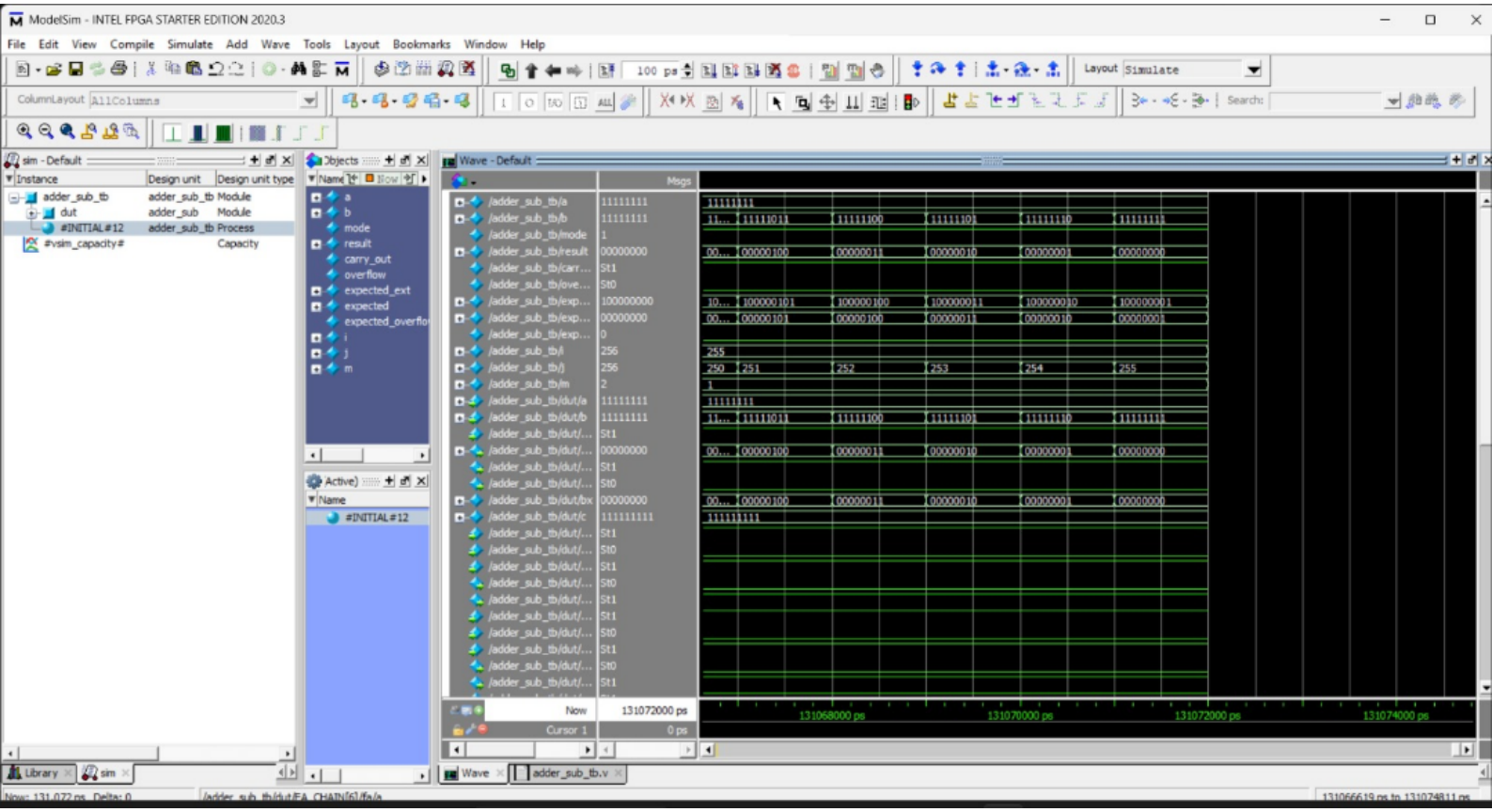

ModelSim - INTEL FPGA STARTER EDITION 2020.3
File Edit View Compile Simulate Add Wave Tools Layout Bookmarks Window Help
sim - Default
Instance
Design unit
Design unit type
fifo_tb
cycle
dut
#ALWAYS#9
#INITIAL#11
#vsim_capacity#
Objects
clk
reset
wr_en
rd_en
data_in
data_out
full
empty
i
Active)
#INITIAL#11
Wave - Default
Msgs
/fifo_tb/clk
/fifo_tb/reset
/fifo_tb/wr_en
/fifo_tb/rd_en
/fifo_tb/data_in
/fifo_tb/data_out
/fifo_tb/full
/fifo_tb/empty
/fifo_tb/i
/fifo_tb/dut/clk
/fifo_tb/dut/reset
/fifo_tb/dut/wr_en
/fifo_tb/dut/rd_en
/fifo_tb/dut/data_in
/fifo_tb/dut/data_out
/fifo_tb/dut/full
/fifo_tb/dut/empty
/fifo_tb/dut/wr_ptr
/fifo_tb/dut/rd_ptr
/fifo_tb/dut/count
Now
380000 ps
Cursor 1
0 ps
80000 ps
100000 ps
120000 ps
140000 ps
160000 ps
180000 ps
Library
Memory List
sim
Wave
fifo_tb.v

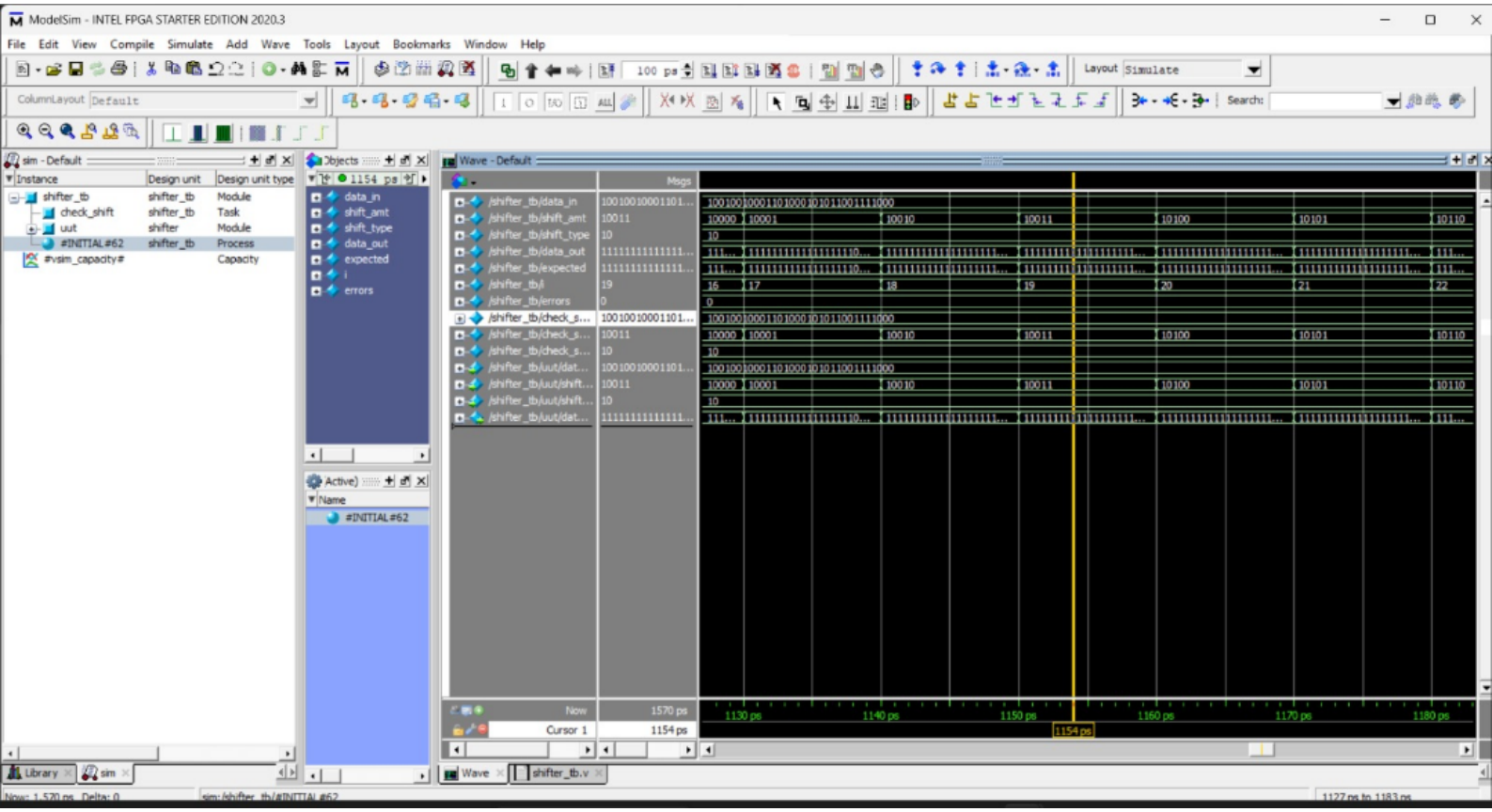

ModelSim - INTEL FPGA STARTER EDITION 2020.3
File Edit View Compile Simulate Add Wave Tools Layout Bookmarks Window Help
sim - Default
Instance Design unit Design unit type
vending_tb vending_tb Module
tick vending_tb Task
dut vending Module
#ALWAYS#7 vending_tb Process
#INITIAL#9 vending_tb Process
#vsim_capacity# Capacity
Objects
clk
reset
coin5
coin10
cancel
dispense
change
Active)
Name
#INITIAL#9
Wave - Default
Msgs
/vending_tb/clk
/vending_tb/reset
/vending_tb/coin5
/vending_tb/coin10
/vending_tb/cancel
/vending_tb/dispense
/vending_tb/change
/vending_tb/dut/clk
/vending_tb/dut/reset
/vending_tb/dut/coin5
sim:/vending_tb/change @ 20574 ps
00000
Now 120000 ps
Cursor 1 0 ps
100 ps 20000 ps 30000 ps 40000 ps 50000 ps 60000 ps 70000 ps
Library sim
Wave vending_tb.v

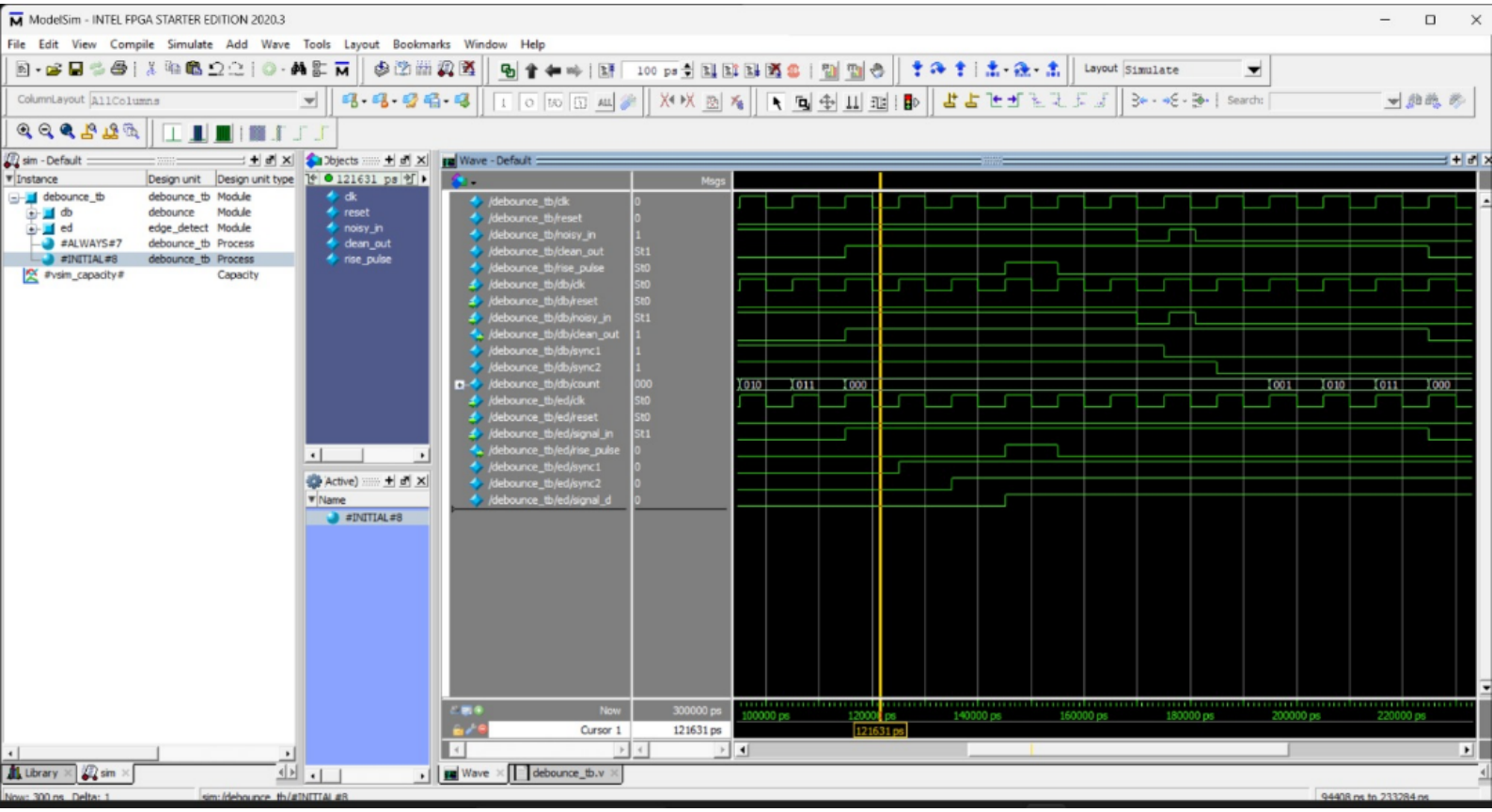

ModelSim - INTEL FPGA STARTER EDITION 2020.3
File Edit View Compile Simulate Add Wave Tools Layout Bookmarks Window Help
sim - Default
Instance Design unit Design unit type
seq_tb seq_tb Module
run_stream seq_tb Task
m seq_moore Module
y seq_mealy Module
#ALWAYS#10 seq_tb Process
#INITIAL#25 seq_tb Process
#vsim_capacity# Capacity
Objects
clk
reset
serial_in
moore_detected
mealy_detected
i
streams
lengths
Active)
#INITIAL#25
Wave - Default
Msgs
/seq_tb/clk
/seq_tb/reset
/seq_tb/serial_in
/seq_tb/moore_det...
/seq_tb/mealy_det...
/seq_tb/i
/seq_tb/run_stream...
/seq_tb/run_stream/n
/seq_tb/run_stream/k
/seq_tb/m/clk
/seq_tb/m/reset
/seq_tb/m/serial_in
/seq_tb/m/detected
/seq_tb/m/state
/seq_tb/m/next_state
/seq_tb/y/clk
/seq_tb/y/reset
/seq_tb/y/serial_in
/seq_tb/y/detected
/seq_tb/y/state
/seq_tb/y/next_state
Now 360000 ps
Cursor 1 0 ps
80000 ps
120000 ps
160000 ps
200000 ps
Library
Memory List
sim
Wave
seq_tb.v

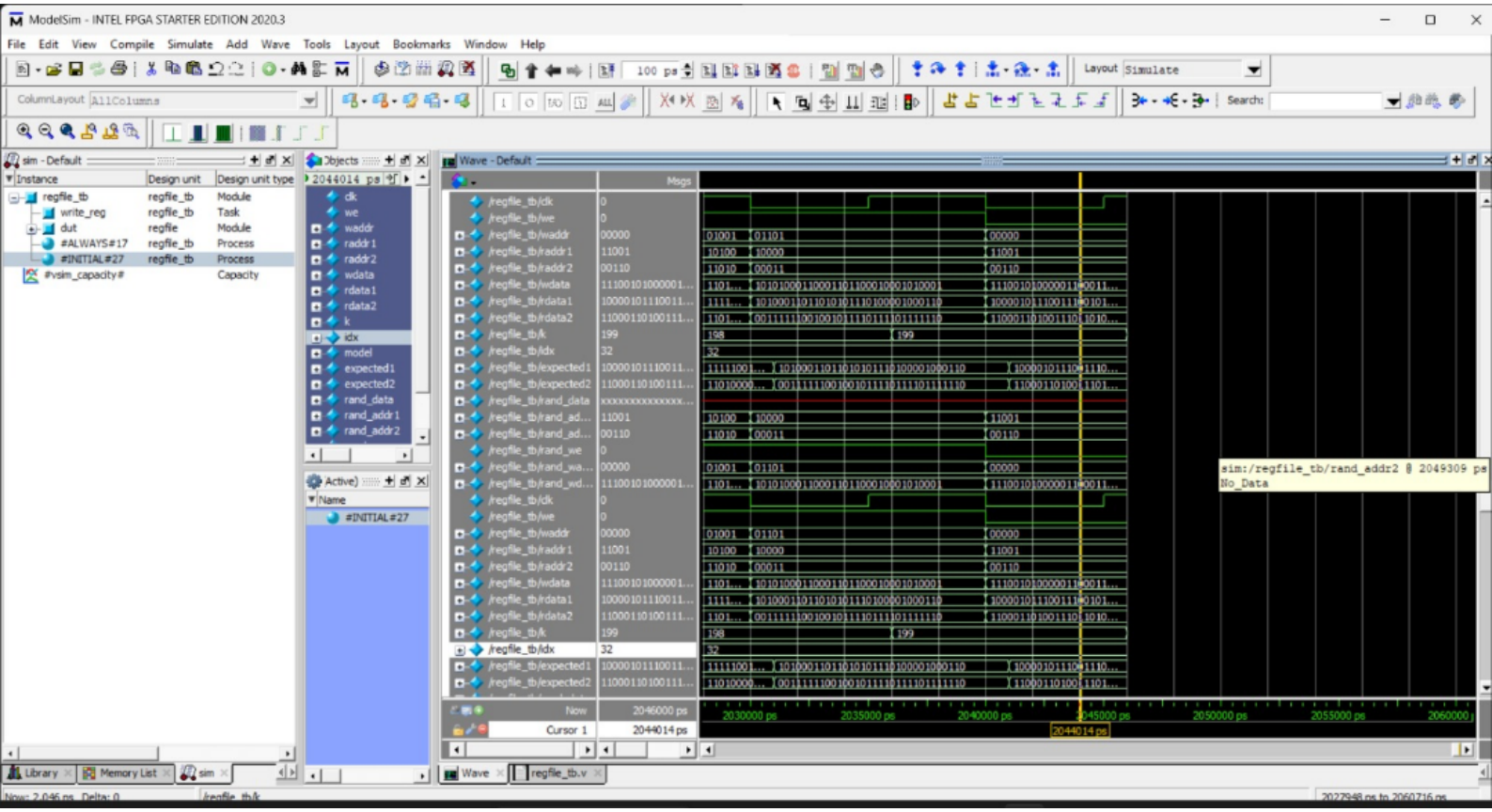

ModelSim - INTEL FPGA STARTER EDITION 2020.3
File Edit View Compile Simulate Add Wave Tools Layout Bookmarks Window Help
sim - Default
Instance
Design unit
Design unit type
encoder_decoder_...
e1
e2
d1
#INITIAL#13
#vsim_capacity#
Objects
in
dec_in
enable
code1
code2
valid1
valid2
dec_out
i
Active)
#INITIAL#13
Wave - Default
Msgs
Now
272000 ps
Cursor 1
0 ps
258000 ps
259000 ps
260000 ps
261000 ps
262000 ps
263000 ps
264000 ps
265000 ps
Wave
encoder_decoder_tb.v
Library
sim

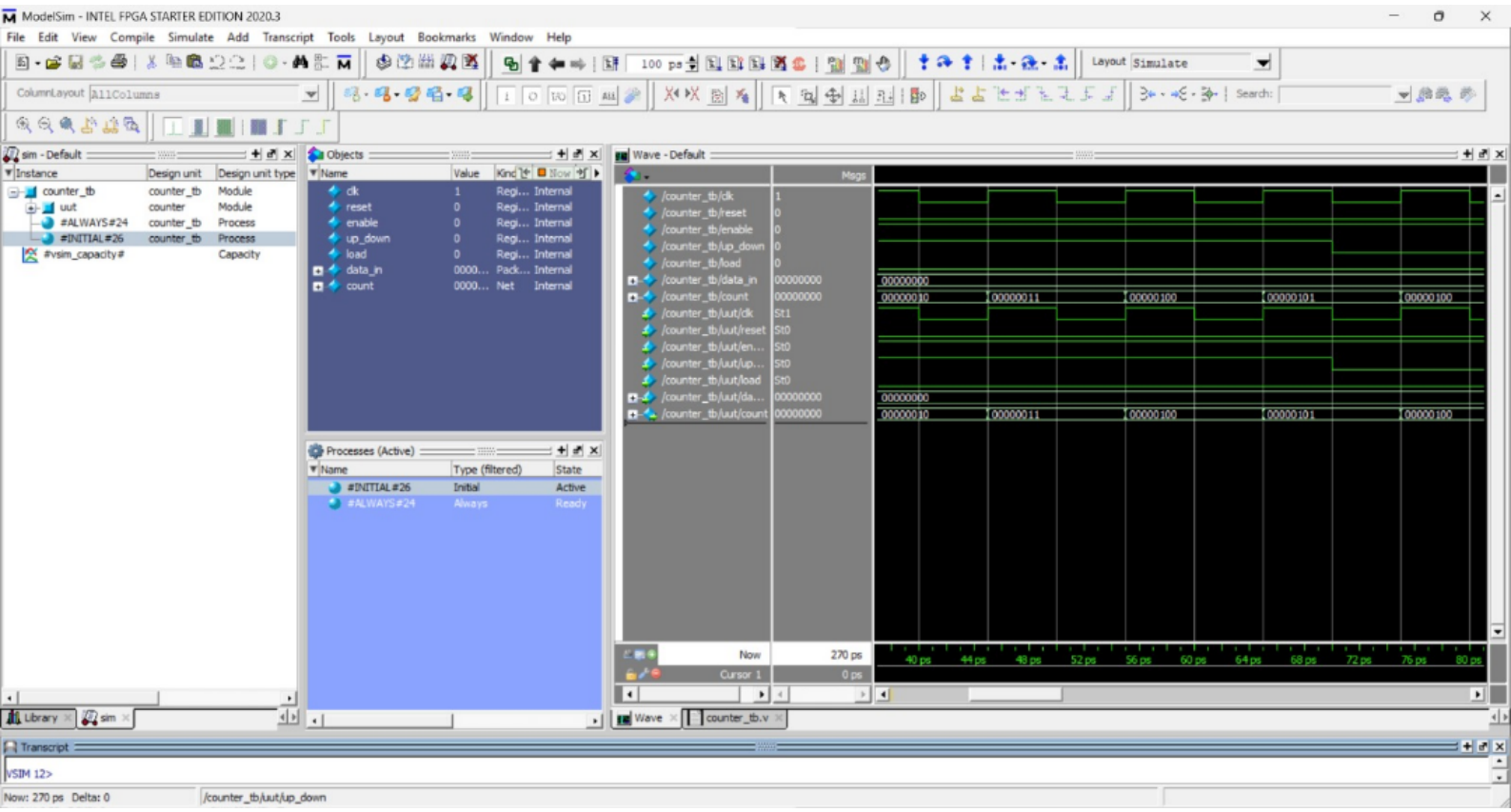

ModelSim - INTEL FPGA STARTER EDITION 2020.3
File Edit View Compile Simulate Add Wave Tools Layout Bookmarks Window Help
sim - Default
Instance
Design unit
Design unit type
shiftreg_lfsr_tb
sr
lf
#ALWAYS#13
#INITIAL#15
#vsim_capacity#
Objects
clk
reset
seed_load
mode
parallel_in
serial_in
seed
q_shift
q_lfsr
seen
i
j
Active)
#INITIAL#15
Wave - Default
Msgs
/shiftreg_lfsr_tb/clk
/shiftreg_lfsr_tb/reset
/shiftreg_lfsr_tb/mode
/shiftreg_lfsr_tb/parallel_in
/shiftreg_lfsr_tb/seed
/shiftreg_lfsr_tb/q_shift
/shiftreg_lfsr_tb/q_lfsr
/shiftreg_lfsr_tb/i
/shiftreg_lfsr_tb/j
Now
2610000 ps
Cursor 1
0 ps
800000 ps
820000 ps
840000 ps
860000 ps
Library
Memory List
sim
Wave
shiftreg_lfsr_tb.v

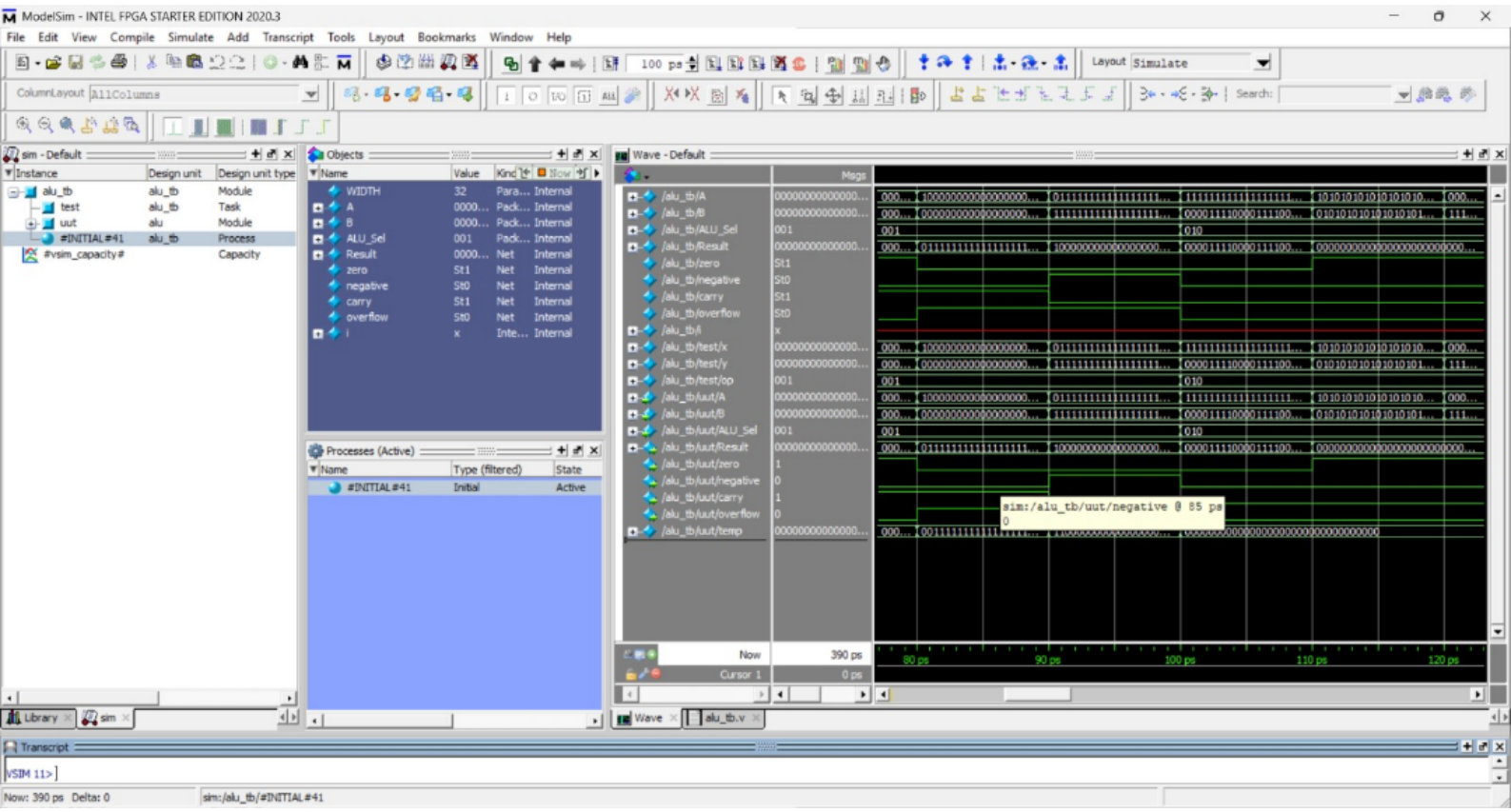

ModelSim - INTEL FPGA STARTER EDITION 2020.3
File Edit View Compile Simulate Add Wave Tools Layout Bookmarks Window Help
sim - Default
Instance
Design unit
Design unit type
adder_sub_tb
dut
#INITIAL#12
#vsim_capacity#
Objects
Wave - Default
Msgs
/adder_sub_tb/a
/adder_sub_tb/b
/adder_sub_tb/mode
/adder_sub_tb/result
/adder_sub_tb/carr...
/adder_sub_tb/ove...
/adder_sub_tb/exp...
/adder_sub_tb/exp...
/adder_sub_tb/exp...
/adder_sub_tb/i
/adder_sub_tb/j
/adder_sub_tb/m
/adder_sub_tb/dut/a
/adder_sub_tb/dut/b
/adder_sub_tb/dut/...
/adder_sub_tb/dut/...
/adder_sub_tb/dut/...
/adder_sub_tb/dut/...
/adder_sub_tb/dut/bx
/adder_sub_tb/dut/c
/adder_sub_tb/dut/...
Now
131072000 ps
Cursor 1
0 ps
131068000 ps
131070000 ps
131072000 ps
131074000 ps
Wave
adder_sub_tb.v


ModelSim - INTEL FPGA STARTER EDITION 2020.3
File Edit View Compile Simulate Add Wave Tools Layout Bookmarks Window Help
sim - Default
Instance Design unit Design unit type
shifter_tb shifter_tb Module
check_shift shifter_tb Task
uut shifter Module
#INITIAL#62 shifter_tb Process
#vsim_capacity# Capacity
Objects
data_in
shift_amt
shift_type
data_out
expected
i
errors
Active)
Name
#INITIAL#62
Wave - Default
Msgs
/shifter_tb/data_in
/shifter_tb/shift_amt
/shifter_tb/shift_type
/shifter_tb/data_out
/shifter_tb/expected
/shifter_tb/i
/shifter_tb/errors
/shifter_tb/check_s...
/shifter_tb/check_s...
/shifter_tb/check_s...
/shifter_tb/uut/dat...
/shifter_tb/uut/shift...
/shifter_tb/uut/shift...
/shifter_tb/uut/dat...
Now 1570 ps
Cursor 1 1154 ps
1130 ps
1140 ps
1150 ps
1160 ps
1170 ps
1180 ps
1154 ps
Library
sim
Wave
shifter_tb.v


ModelSim - INTEL FPGA STARTER EDITION 2020.3
File Edit View Compile Simulate Add Wave Tools Layout Bookmarks Window Help
sim - Default
Instance Design unit Design unit type
debounce_tb debounce_tb Module
db debounce Module
ed edge_detect Module
#ALWAYS#7 debounce_tb Process
#INITIAL#8 debounce_tb Process
#vsim_capacity# Capacity
Objects
121631 ps
clk
reset
noisy_in
clean_out
rise_pulse
Active)
Name
#INITIAL#8
Wave - Default
Msgs
/debounce_tb/clk
/debounce_tb/reset
/debounce_tb/noisy_in
/debounce_tb/clean_out
/debounce_tb/rise_pulse
/debounce_tb/db/clk
/debounce_tb/db/reset
/debounce_tb/db/noisy_in
/debounce_tb/db/clean_out
/debounce_tb/db/sync1
/debounce_tb/db/sync2
/debounce_tb/db/count
/debounce_tb/ed/clk
/debounce_tb/ed/reset
/debounce_tb/ed/signal_in
/debounce_tb/ed/rise_pulse
/debounce_tb/ed/sync1
/debounce_tb/ed/sync2
/debounce_tb/ed/signal_d
010
011
000
001
010
011
000
Now
300000 ps
Cursor 1
121631 ps
100000 ps
120000 ps
140000 ps
160000 ps
180000 ps
200000 ps
220000 ps
121631 ps
Library
sim
Wave
debounce_tb.v


ModelSim - INTEL FPGA STARTER EDITION 2020.3
File Edit View Compile Simulate Add Wave Tools Layout Bookmarks Window Help
sim - Default
Instance
Design unit
Design unit type
regfile_tb
write_reg
dut
#ALWAYS#17
#INITIAL#27
#vsim_capacity#
Objects
clk
we
waddr
raddr1
raddr2
wdata
rdata1
rdata2
k
idx
model
expected1
expected2
rand_data
rand_addr1
rand_addr2
Wave - Default
Msgs
/regfile_tb/clk
/regfile_tb/we
/regfile_tb/waddr
/regfile_tb/raddr1
/regfile_tb/raddr2
/regfile_tb/wdata
/regfile_tb/rdata1
/regfile_tb/rdata2
/regfile_tb/k
/regfile_tb/idx
/regfile_tb/expected1
/regfile_tb/expected2
/regfile_tb/rand_data
/regfile_tb/rand_we
/regfile_tb/rand_wa...
/regfile_tb/rand_wd...
#sim:/regfile_tb/rand_addr2 @ 2049309 ps
#No_Data
Now
Cursor 1
2046000 ps
2044014 ps
2030000 ps
2035000 ps
2040000 ps
2045000 ps
2050000 ps
2055000 ps
2060000 ps
Library
Memory List
sim
Wave
regfile_tb.v


ModelSim - INTEL FPGA STARTER EDITION 2020.3
File Edit View Compile Simulate Add Transcript Tools Layout Bookmarks Window Help
sim - Default
Instance Design unit Design unit type
counter_tb counter_tb Module
uut counter Module
#ALWAYS#24 counter_tb Process
#INITIAL#26 counter_tb Process
#vsim_capacity# Capacity
Objects
Name Value Kind Now
clk 1 Regi... Internal
reset 0 Regi... Internal
enable 0 Regi... Internal
up_down 0 Regi... Internal
load 0 Regi... Internal
data_in 0000... Pack... Internal
count 0000... Net Internal
Processes (Active)
Name Type (filtered) State
#INITIAL#26 Initial Active
#ALWAYS#24 Always Ready
Wave - Default
Msgs
/counter_tb/clk
/counter_tb/reset
/counter_tb/enable
/counter_tb/up_down
/counter_tb/load
/counter_tb/data_in
/counter_tb/count
/counter_tb/uut/clk
/counter_tb/uut/reset
/counter_tb/uut/en...
/counter_tb/uut/up...
/counter_tb/uut/load
/counter_tb/uut/da...
/counter_tb/uut/count
00000010
00000011
00000100
00000101
Now 270 ps
Cursor 1 0 ps
40 ps 44 ps 48 ps 52 ps 56 ps 60 ps 64 ps 68 ps 72 ps 76 ps 80 ps
Library sim
Wave counter_tb.v
Transcript
VSIM 12>
Now: 270 ps Delta: 0
/counter_tb/uut/up_down


ModelSim - INTEL FPGA STARTER EDITION 2020.3
File Edit View Compile Simulate Add Transcript Tools Layout Bookmarks Window Help
sim - Default
Instance Design unit Design unit type
alu_tb alu_tb Module
test alu_tb Task
uut alu Module
#INITIAL#41 alu_tb Process
#vsim_capacity# Capacity
Objects
Name Value Kind
WIDTH 32 Para... Internal
A 0000... Pack... Internal
B 0000... Pack... Internal
ALU_Sel 001 Pack... Internal
Result 0000... Net Internal
zero St1 Net Internal
negative St0 Net Internal
carry St1 Net Internal
overflow St0 Net Internal
i x Inte... Internal
Processes (Active)
Name Type (filtered) State
#INITIAL#41 Initial Active
Wave - Default
Msgs
/alu_tb/A
/alu_tb/B
/alu_tb/ALU_Sel
/alu_tb/Result
/alu_tb/zero
/alu_tb/negative
/alu_tb/carry
/alu_tb/overflow
/alu_tb/i
/alu_tb/test/x
/alu_tb/test/y
/alu_tb/test/op
/alu_tb/uut/A
/alu_tb/uut/B
/alu_tb/uut/ALU_Sel
/alu_tb/uut/Result
/alu_tb/uut/zero
/alu_tb/uut/negative
/alu_tb/uut/carry
/alu_tb/uut/overflow
/alu_tb/uut/temp
sim:/alu_tb/uut/negative @ 85 ps
Now 390 ps
Cursor 1 0 ps
80 ps 90 ps 100 ps 110 ps 120 ps
Transcript
VSIM 11>
Now: 390 ps Delta: 0
sim:/alu_tb/#INITIAL#41